

INTRODUCTION TO ECONOMICS

MGT404

Trade

Yale

Lecture Plan: Trade and Applications

1. Introduction to Trade

- Absolute and Comparative Advantage
- Gains from Trade

2. Trade Policy Analysis

- Autarky versus free trade
- Imports and exports with the rest of the world
- Tariffs and subsidies

3. Applications and Discussion

- We've been down the road of tariffs before.
- Tennis Balls??
- Washing machines were an early target.

Introduction: What is Trade?

“Trade” as a General Idea

- “Trade” is often synonymous with *international trade*, but in fact is a more general idea. Trade occurs between firms, between individuals, within teams, etc.
- Common theme: we are studying trade that is voluntary: both parties opt-in to the transaction. Logically, it seems both are better off from trading.

“Trade” Transaction

Labor for a Wage

Going out to dinner

The USA importing Kiwis
from New Zealand

Implicit Acknowledgement

- Firm prefers paying wage for labor than any other input; employee prefers wage to outside options.
- It would be costly for me to prepare a meal, and the restaurant is happy to provide it at the menu prices.
- Americans like Kiwis at the price NZ is willing to sell; NZ is willing to part with Kiwis at that price.

Introduction: Gains from Trade

How does each party fare from this transaction?

Labor for a Wage

- Firm: better off! Without labor, would either produce less or earn less.
- Employee: better off! Without job, worse off.

Going out to dinner

- Firm: better off! We build in a markup
 - Employees: see above
- Me: I wouldn't eat out if I'd trade the price for cooking myself, so I'm better off.

The USA importing Kiwis from New Zealand

- Americans: better off! We have kiwis!
- NZ: better off! Happy to trade away excess kiwis for US dollars!



When two parties engage in voluntary trade, it is said that there are “**gains from trade**”: that is, the increase in surplus from not having the opportunity to trade.

Article Discussion: Blinder's Intro to Trade

Three truths about trade:

1. Trade deficits and surpluses are not profits or losses

- The US will always have a trade deficit in bananas.
- That is not the result of “unfair” trade. It is a result of bananas being delicious.

2. The USA imports a lot of capital

- Luckily, the US is able to borrow very freely from the rest of the world. We do, and we consume lots of imported goods.

3. It makes sense to import things someone else can make more efficiently

- The USA is really good at making aerospace, IP, software, entertainment goods, agriculture, pharma. It should make lots of these things.
- The USA should then import things that other places are better at making.



Trade is a tough topic and it is easy to draw the wrong conclusions. Today, we will walk through trade and trade policies to understand what they do.

Introduction: Comparative Advantage



“It's reported that when John Lennon was asked whether Ringo Starr was the best drummer in the world, he responded that Ringo wasn't even the best drummer in the Beatles.”

Introduction: Advantages

Let us now consider two forms of “advantage” in production. Consider this hypothetical:

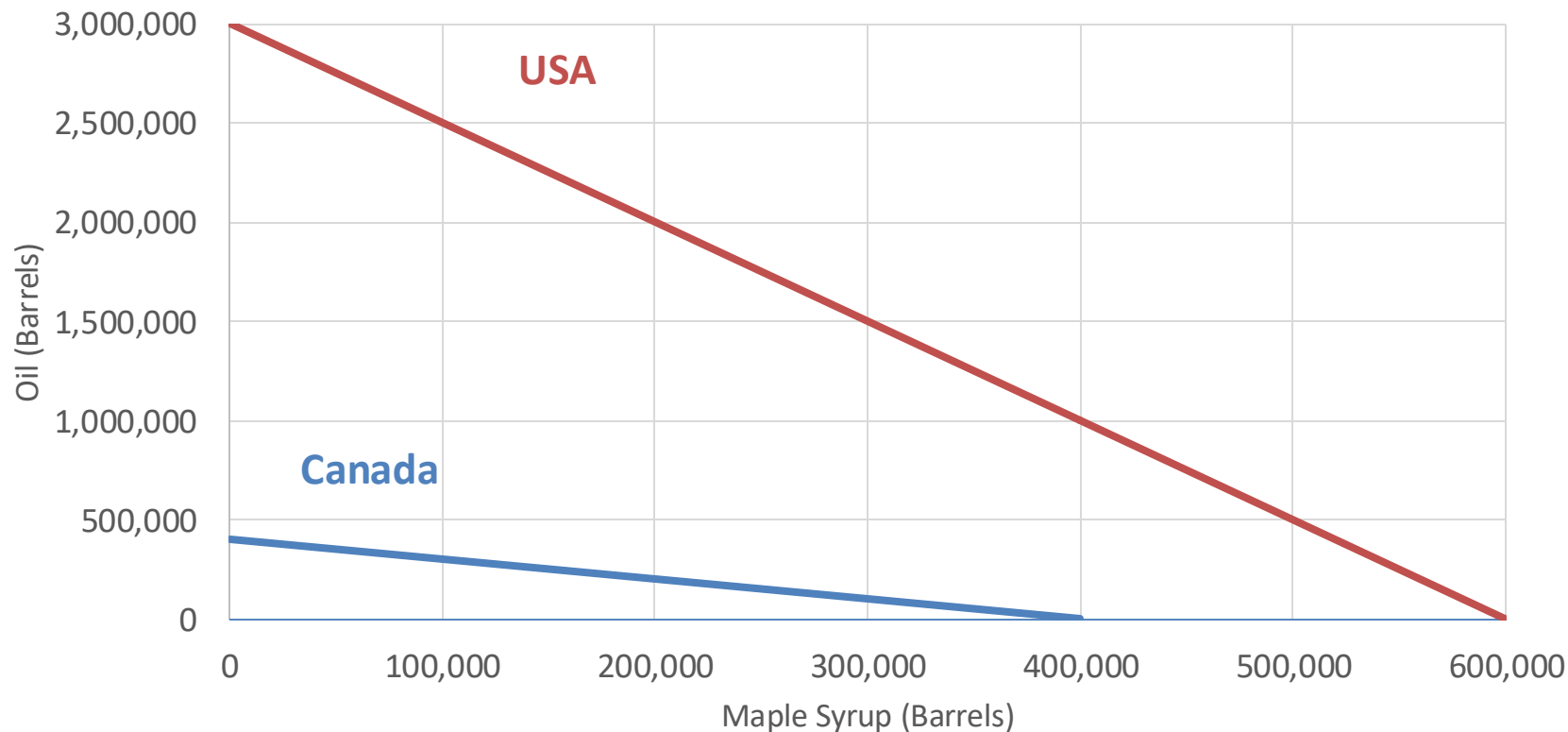
	Canada	USA
Oil & Gas	• Max Production: 400,000 barrels/day equiv.	• Max production: 3,000,000 barrels/day equiv.
Maple Syrup	• Max Production: 400,000 barrels/day equiv.	• Max production: 600,000 barrels/day equiv.
Produce Both	• Can trade off one barrel of oil for one barrel of maple syrup	• 1 barrel of Maple Syrup per 5 barrels of oil/gas



In this example the USA has an **absolute advantage (more outputs per input)** in production of oil & gas and maple syrup. Given one day (input), the USA can produce more outputs of either.

Question: Why would the USA ever engage in trade with Canada for either Oil & Gas or Maple Syrup?

Production Possibility Frontiers: USA and Canada



In the absence of trade, each country could pick a point on their line to determine how much to produce.

Introduction: Advantages – the insight:

Let us now consider two forms of “advantage” in production. Consider:

	Canada	USA
Tradeoff:	• 1 barrel of Maple Syrup per 1 barrel of oil/gas	• 1 barrel of Maple Syrup per 5 barrels of oil/gas

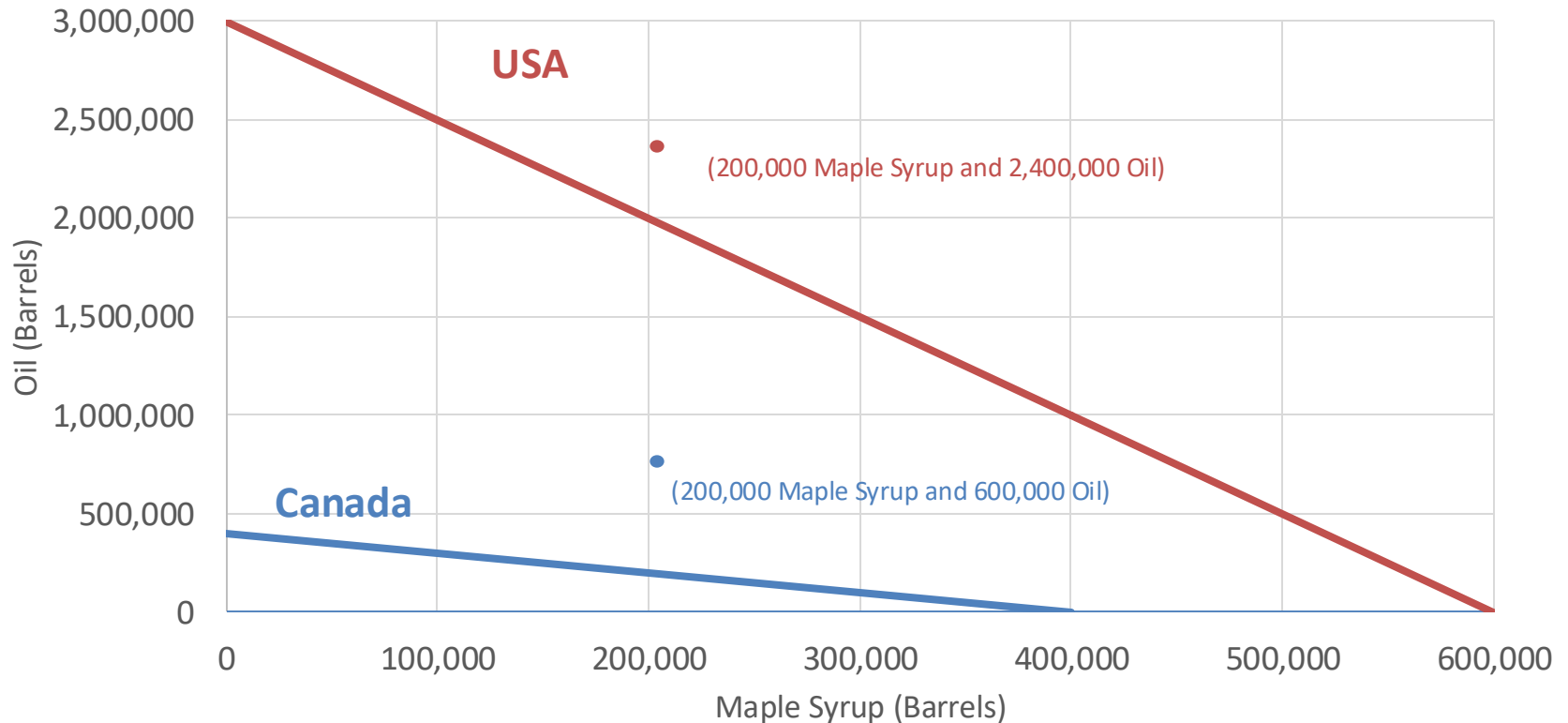
Notice that the USA has a high “opportunity cost” of producing Maple Syrup. A barrel of Maple Syrup displaces 5 barrels of oil/gas, instead of only 1 in Canada.

For every barrel of Maple Syrup demanded in the US, instead of trading five barrels of oil/gas within the US, instead trade 1-5 barrels with Canada.

This insight is called “**comparative advantage**”. How is this useful?
Let’s suppose the two countries agree on an exchange rate: 3 barrels of Oil for 1 barrel of Maple Syrup.

Production Possibilities, USA and Canada

Suppose Canada specializes in Maple Syrup and the USA in Oil, and there is a 1:3 exchange rate for Oil (1 bbl of Maple Syrup for 3 bbls of Oil)



Trade expands the frontier: both countries can be better off!

This idea has a lot of power

“Trade” Concepts:

- “**Absolute Advantage**”: I can produce more output with the same or fewer inputs.
- “**Comparative Advantage**”: I can produce one output more efficiently than you *in terms of other outputs*.
- “**Gains from Specialization**”: We are both better off if we **specialize** and trade instead of acting on our own. Win-win!



Trade is a very general idea. You will explore the big-picture implications in other classes. In this class, let's explore the first-order implications and explore some different policy tools.

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Basic Model: “Small Open Economy” model of trade

Setting

Suppose that at first, the country does not trade with anyone outside its borders. This situation is referred to as “**autarky**”. Then, we are in the standard setup we have been looking at all along: there is a domestic demand curve, a domestic supply curve, and an equilibrium price, P^A . It will then open up to trade.

Assumptions

“Small”

“Open”

Comments

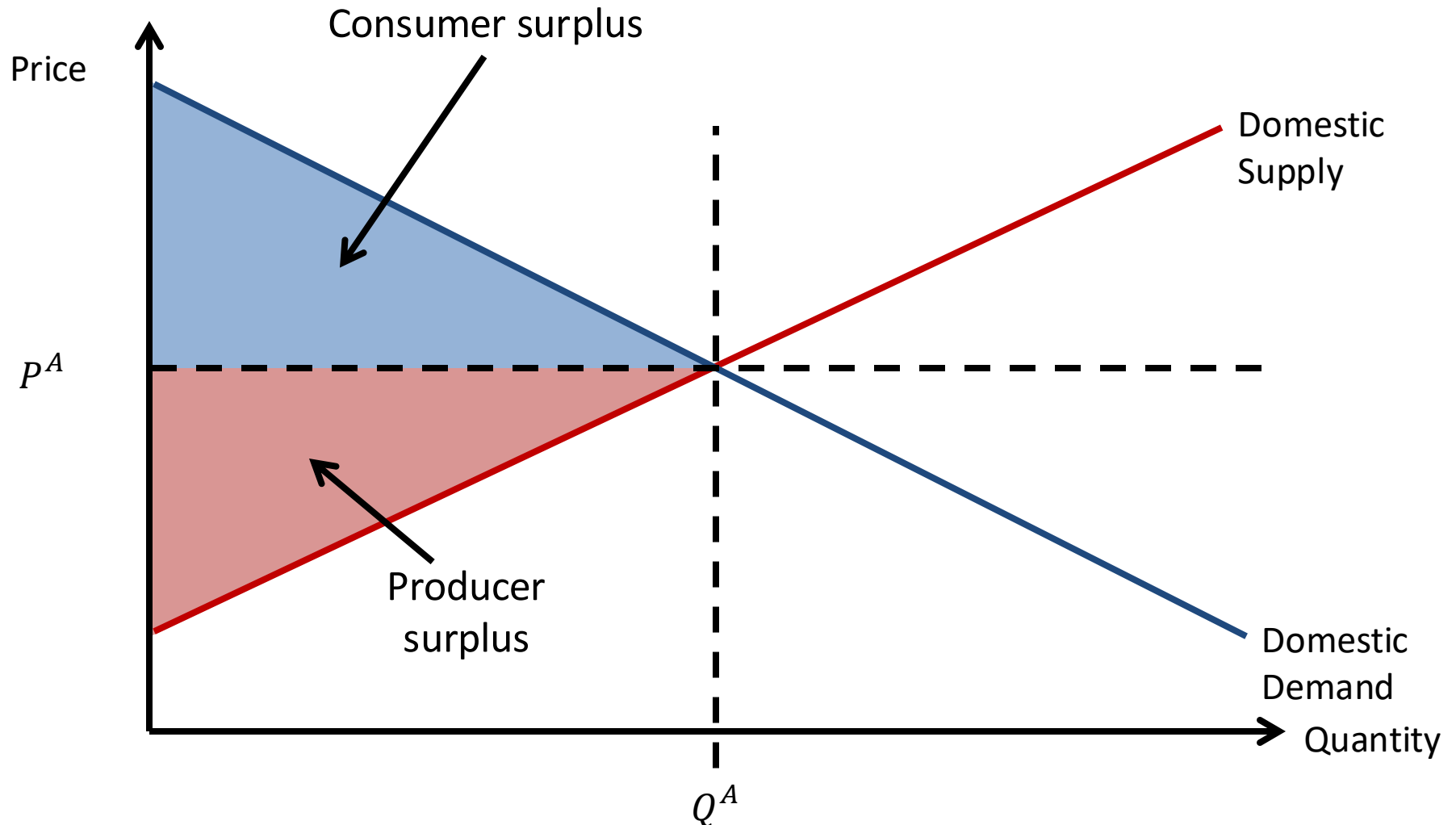
- Key idea is that this “country” doesn’t affect global prices in the market for this good
- This industry is not important to the overall economy
- This country is not charging tariffs or offering subsidies to producers.

When the country opens up to the world, there is a world price of P^W .

First possibility: $P^W > P^A$ (Product examples?)

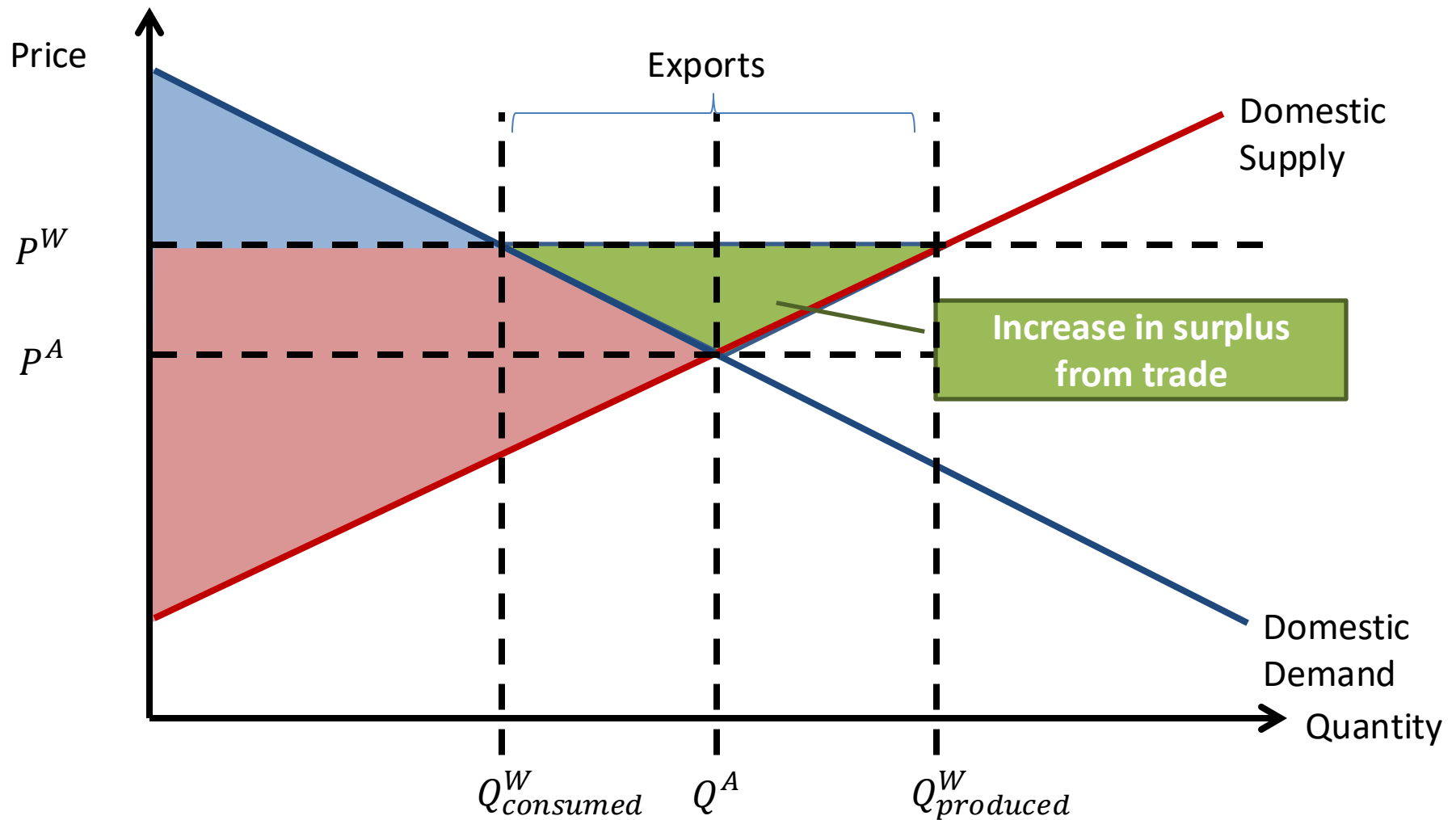
Second possibility: $P^W < P^A$ (Product examples?)

Consider Autarky: Our Standard Market Analysis



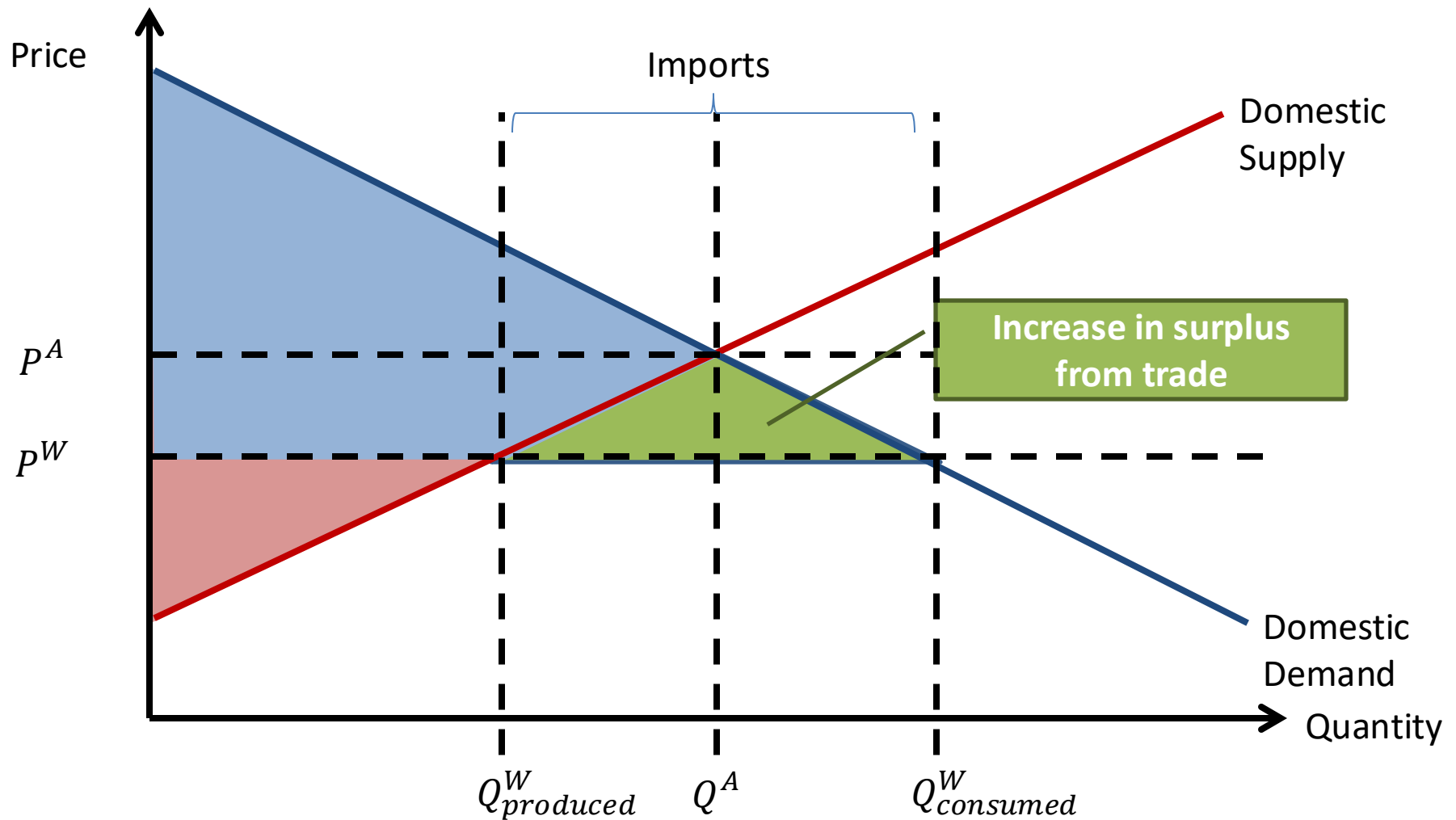
Autarky equilibrium: domestic supply = domestic demand

Now Consider $P^W > P^A$:



Total domestic surplus increases, even though domestic consumer surplus falls.

Now Consider $P^W < P^A$:



Total domestic surplus increases, even though domestic producer surplus falls.

Summary

Case when opening up to trade:

$$P^W > P^A$$

Outcome

- Country becomes an exporter
- Producers do better, consumers worse, **net effect is positive.**

$$P^W < P^A$$

- Country becomes an importer
- Producers do worse, consumers better, **net effect is positive.**



... Why is this controversial?

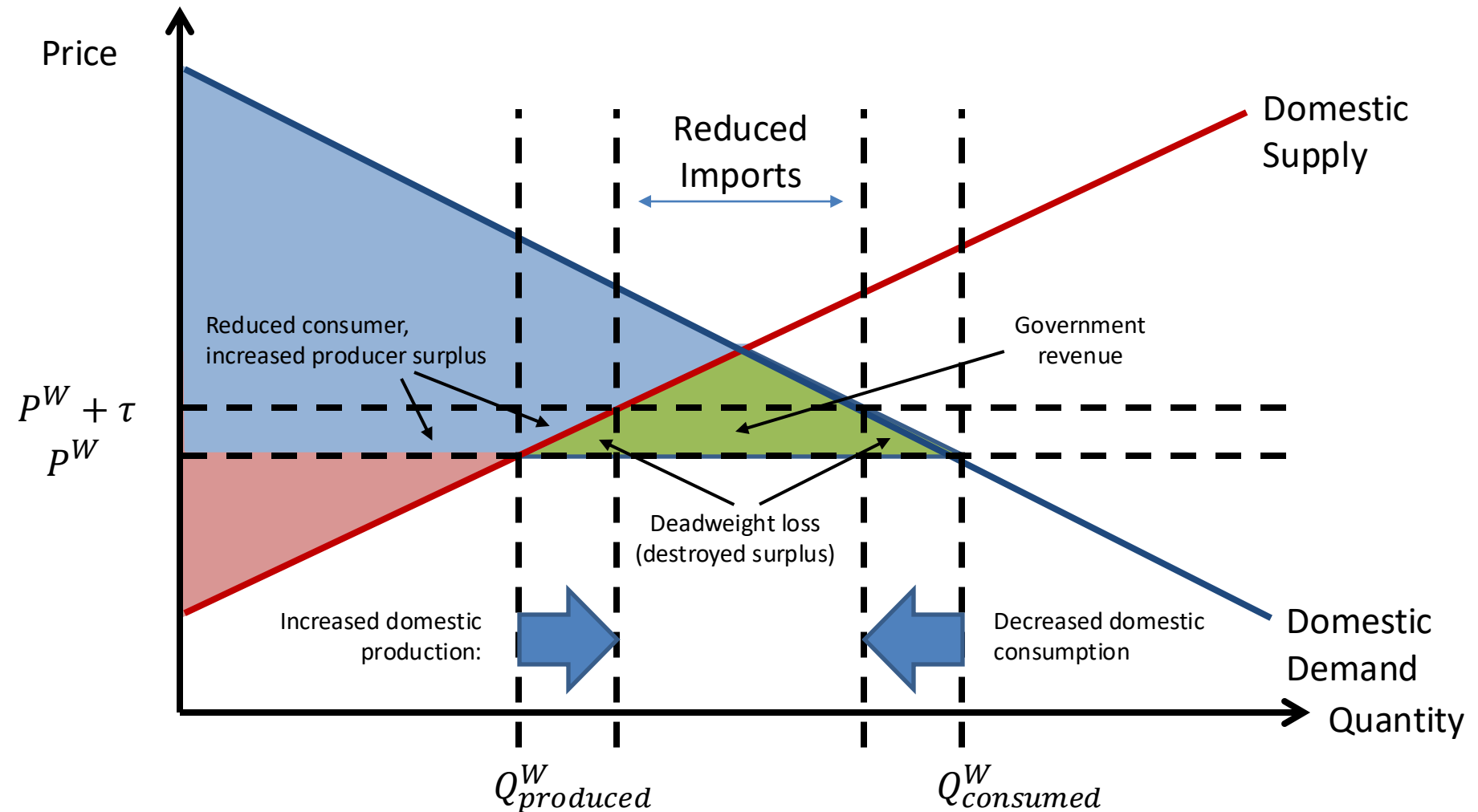
Trade policies create winners and losers. Sometimes the gains are dispersed over a large group (e.g. a small price decrease for millions of consumers) while the costs can be concentrated (hundred of individual jobs lost). Ideally, trade policies that increase total surplus are combined with domestic policies that redistribute gains more evenly.

Trade Policy Tools

Policy Tool:	Description
1. Import Tariff	This is a tax on imports, but not on domestically produced versions of a good.
2. Export Subsidy	Give a transfer (tax credit) for producers that export their products.
3. Export Tax	Tax exports to shield domestic consumers from world trade
4. Quota	Limit the quantity of a good that can be imported from abroad
5. Voluntary Restraints	Voluntary Export Restraints (VERs) are self-imposed quotas.

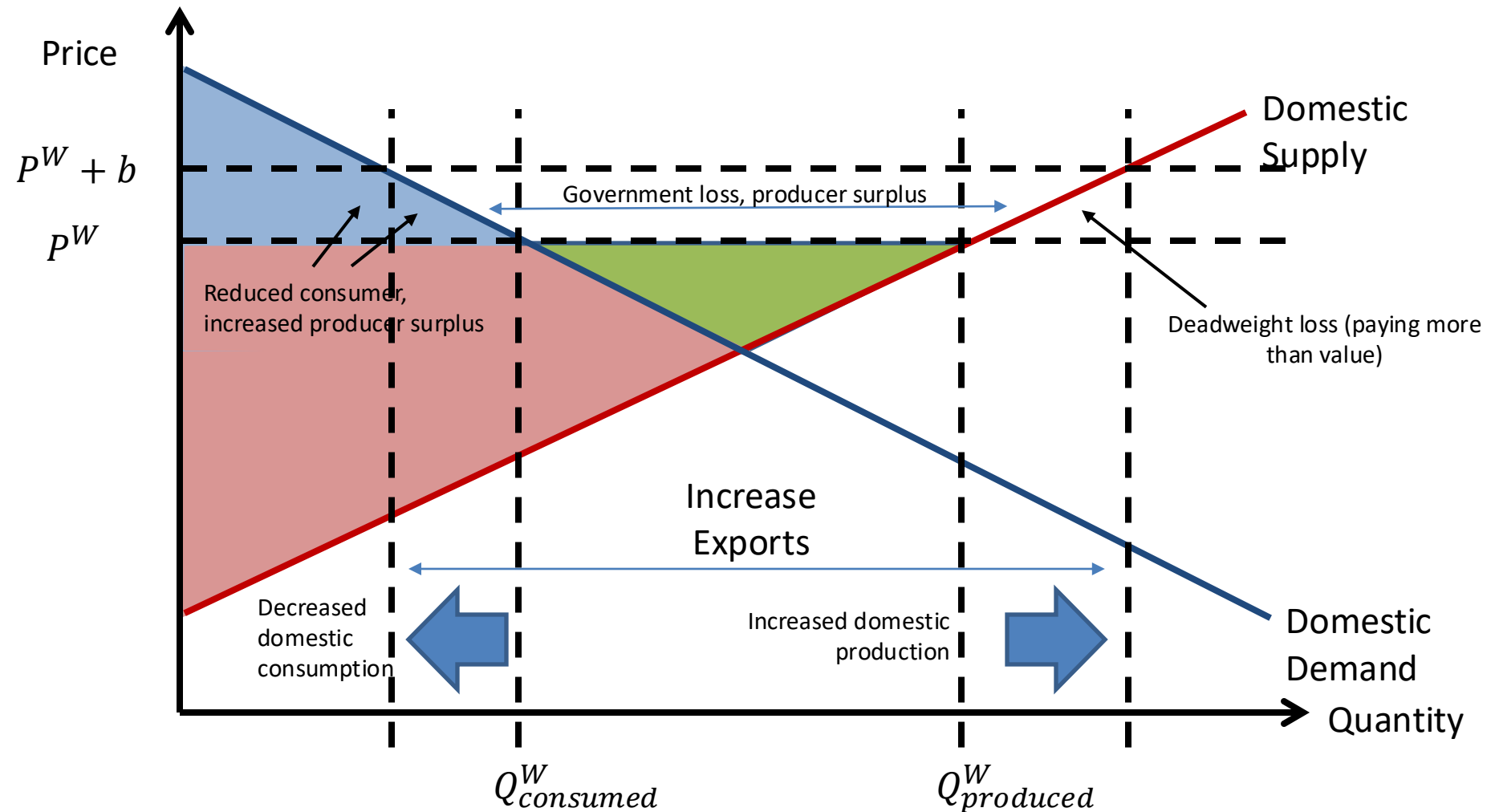
1. Import Tariff

An import tariff is a “tax”, τ , on the world price of a good, P^W .



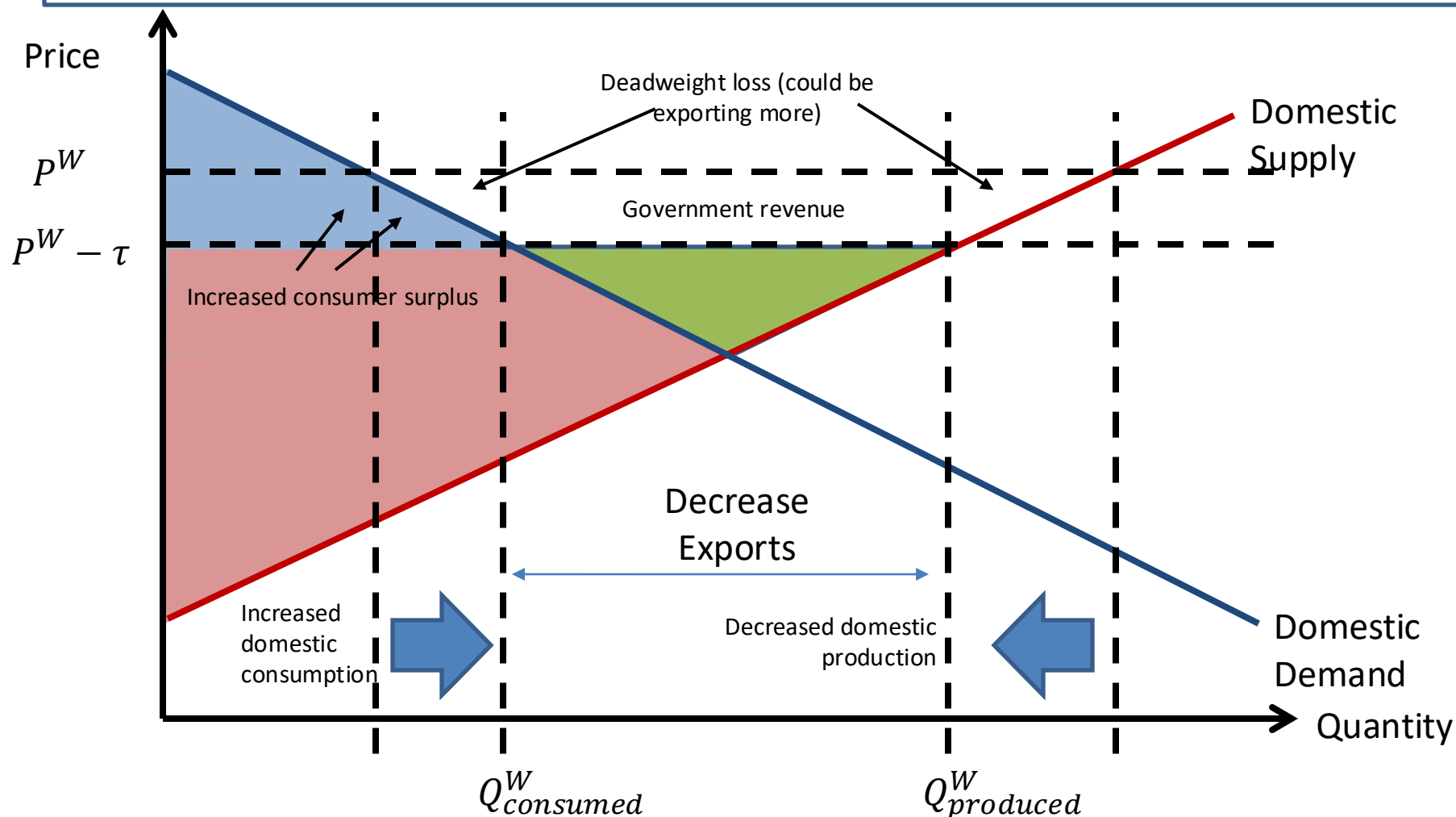
2. Export Subsidy:

An export subsidy is effectively a “bonus”, b , on exporters, on the world price of a good.



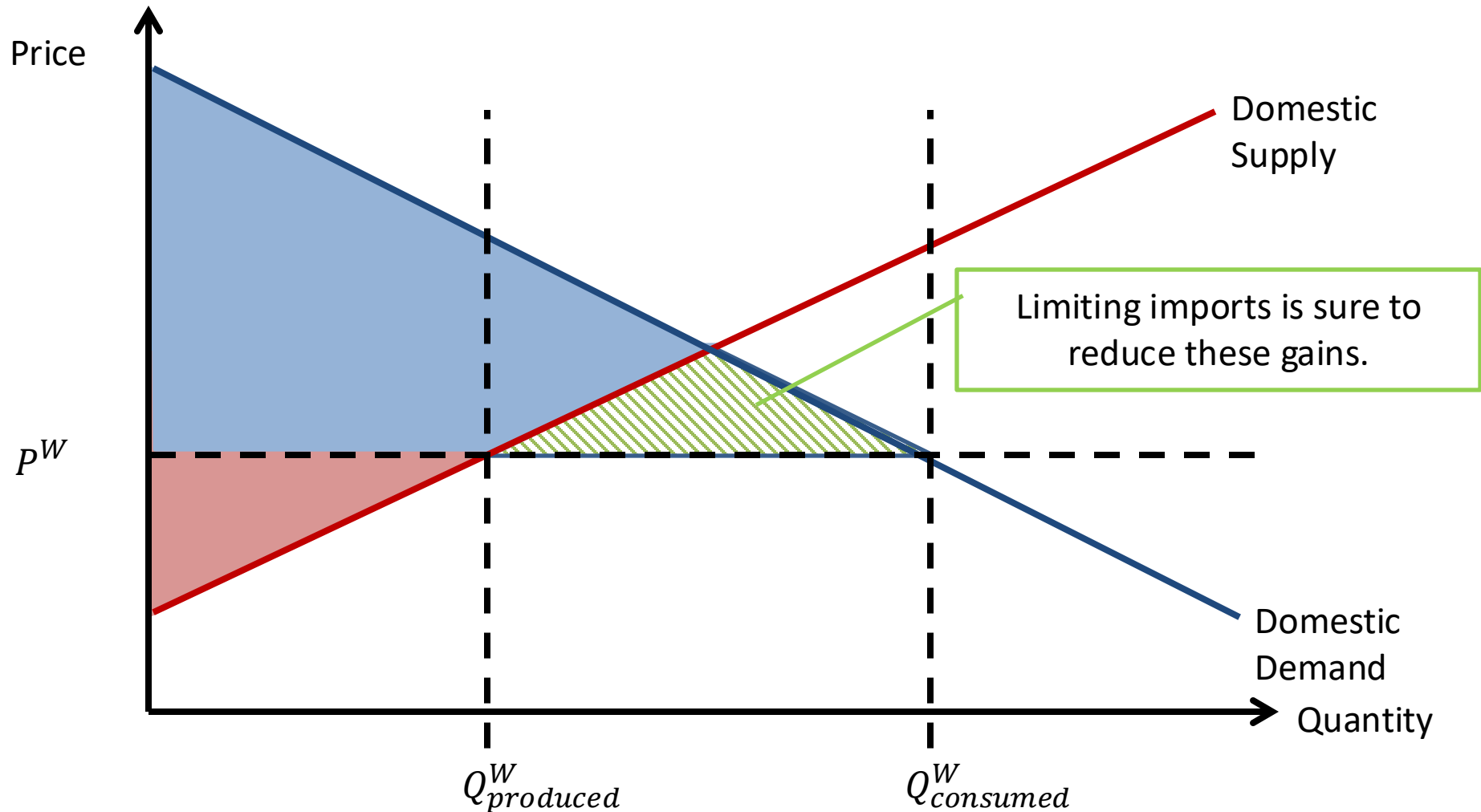
3. Export Tax: Opposite of Export Subsidy (Lowers World Price)

An export tax effectively lowers the world price for exporters. This favors domestic consumers.



4. Quota: Maximum level of imports

If a quota limits imports from their free trade level, we would expect **upward pressure on price**, without generating the government revenue of a tariff (unless quotas are purchased).



Trade Policy Tools

Policy Tool:	Outcome
1. Import Tariff	• Decreases total surplus: benefits domestic producers and raises gov revenue, but harms consumers and creates DWL.
2. Export Subsidy	• Decreases total surplus: raises prices to domestic buyers and costs gov money; benefits producers and creates DWL.
3. Export Tax	• Decreases total surplus: favors domestic consumers (which may be other industries as opposed to individuals)
4. Quota	• Decreases total surplus: doesn't raise government revenue, but raises consumer prices.
5. Voluntary Restraints	• Decreases total surplus: A tactic to head off tariffs.



There don't seem to be any easy wins here.

Sure enough, most trade policy tools are seen as having more costs than benefits.

Free trade will “**maximize the size of the pie**”; however it creates **winners and losers**. Redistribution (dividing the pie) needs to take that into account.

Reasons in Favor of Trade Restraints

The previous slides claimed that trade was generally positive for total surplus. What arguments might economists make in favor of trade restraints?

Reason

Rationale

Domestic subsidies

- If a trading partner engages in domestic subsidies, then free trade could be a bad idea for producers as the playing field will not be level (even though domestic consumers benefit...).

“National Security”

- Suppose a domestic industry has a minimum efficient scale to operate. For example, if the US doesn’t produce 10M tons of steel per year, steel industry shuts down. Then, protect domestic industry. (Note: this is not the case).

Asymmetric Policies

- China enacted rules requiring joint ventures for foreign partners that included technology exchange.
 - These rules seem to have given away trade secrets from US firms to their Chinese counterparts.

Trade negotiations can also be costly. People don’t outsource many aspects of their life because finding a “trade partner” is costly.

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- Tennis Balls
- Busy Baby Mat

Opinions

Trump's proposed tennis ball tariff represents a grand slam of terrible trade policy



A ball shagger watches from the sidelines at the Citi Open Tennis Tournament at the Rock Creek Park Tennis Center in Washington on Aug. 2, 2017. (Calla Kessler/The Washington Post)



By **Catherine Rampell**
Columnist
June 27

Want to know how ill-conceived President Trump's proposed China tariffs are? Look to the tennis ball tariff.

It represents a sort of grand slam of terrible trade policy.

Head Penn Racquet Sports, a division of Colorado-based Head USA Inc., makes the best-selling tennis ball in the United States. It sells about 65 million balls under its "Penn" brand each year, representing nearly two-thirds of all branded tennis ball sales in the United States.

- (Opinion piece, know to check details)
- Many US firms import items from China (or other low-cost manufacturing countries)
- How hard is it to target trade measures?

Opinions

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- How hard is it to target trade measures?

Tennis Balls and Trade Wars

Goal of Tariffs

- Help American Business compete

Result of Tariffs

- American business losing out
- Chinese businesses thriving



Using trade policy for diplomacy is difficult.
It is easy to have unintended consequences.
(Global supply chains are difficult!)

What about Busy Baby?

Good Company » Growth Studio

Busy Baby's Tariff Pivot: Going Global

When tariffs hit, Busy Baby pivoted. Here's how the company turned shipping headaches into a springboard for global expansion.

By: *Miranda Fraraccio*, Contributor

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Beth Benike and Eric Fynbo, owners of Busy Baby. — Busy Baby

Summary

Today's Recap

- Voluntary trade is generally positive: it can produce winners and losers, but total surplus increases.
- Even if a firm or nation has an absolute advantage in production, it can be better off trading due to comparative advantage.
- Interventions in trade such as tariffs tend to be suspect.

Next Lecture: Producing in a perfectly competitive industry

- What is perfect competition?
- What decisions can a firm make?
 - Choosing optimal quantity
 - Choosing whether to operate or not

Appendix – Practice Trade Problem

Trade Practice Problem

Page 1: questions. Pages 2-3 solutions. Print Page 1 to work through, only check solutions afterwards.

The demand for pineapples in Belize is given by:

$$Q_D = 20 - P$$

And the supply there is given by

$$Q_S = 4P - 5$$

Assume Belize is “small” in that it cannot affect the world price of pineapples.

1. What are the autarky (no trade) price and quantity of pineapples in Belize?
2. Suppose free trade prevails and Belize exports 25 pineapples. What must the world price of pineapples be? (Hint: if you're unsure how to begin, draw the graph. Recall that when we plot these functions, we write them as $P = \dots$)

-
3. Now Belize places an export subsidy of 5 on each unit of pineapples exported. What are exports now?

 4. What is the welfare loss or gain associated with the subsidy?

Advice: Attempt questions and discuss with learning team before ever looking at solutions.

(Always attempt on your own before looking at solutions)

1. Solve for the equilibrium by setting demand = supply

$$\begin{aligned}Q_D &= Q_S \\20 - P &= 4P - 5 \\25 &= 5P \\P^* &= \$5 \\Q^* &= 15\end{aligned}$$

2. Belize is exporting 25 pineapples. This means that the quantity supplied must be the quantity demanded domestically at the world price + 25.

$$\begin{aligned}Q_S(P^W) &= Q_D(P^W) + 25 \\4P^W - 5 &= 20 - P^W + 25 \\5P^W &= 50 \\P^W &= \$10\end{aligned}$$

So the world price is \$10. It is easy to now check that $Q_D(10) = 10$, $Q_S(10) = 35$, and so the country must be exporting 25 pineapples.

3. This export subsidy has the effect of increasing the world price to \$15. At that price, we have

$$\begin{aligned}Q_S(15) &= 55 \\Q_D(15) &= 5\end{aligned}$$

Producers would like to supply 55 units at a world price of \$10 + an export subsidy of \$5. Since they only get the export subsidy if they export, any domestic buyers would have to agree to pay \$15 as well (otherwise the producer would export at the world price of \$10 and receive the \$5 export subsidy).

As a result, only 5 units are demanded domestically, and so 50 units are exported.

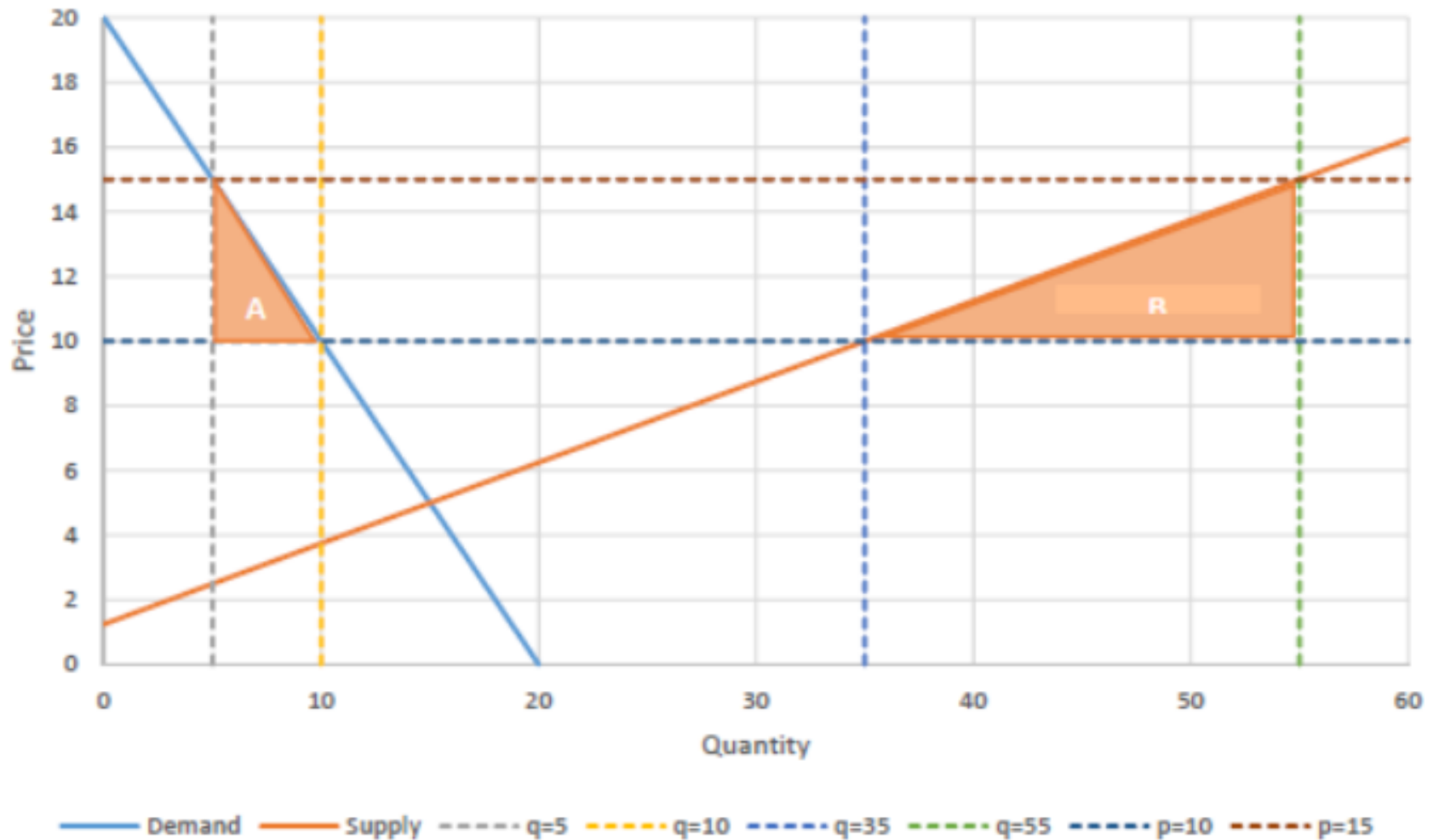
4. There are two deadweight loss regions created by this subsidy (see diagram on following page). The triangles have a combined area of 62.5, so the welfare loss is 62.5.

$$\frac{1}{2}(5)(5) + \frac{1}{2}(20)(5) = \frac{1}{2}(25) + \frac{1}{2}(100) = 62.5$$

Area A: Before subsidy: consumer surplus. After subsidy: producer surplus instead of consumer surplus, but also government expenditure

Area B: Before subsidy: nothing. After subsidy: government expenditure (note that suppliers are not capturing it as producer surplus).

Belize - Free Trade + Export Subsidy



Why Trade is Useful

No Trade

- Suppose both countries dedicate half of their resources to each output.
- Canada would produce 200,000 barrels each of oil and maple syrup
- The USA would produce 1.5M barrels of oil, and 300,000 of Maple Syrup
- Total production is 1.7M barrels of oil and 500,000 barrels of maple syrup

Trade

- If we allow for trade, the two countries could specialize (Canada in maple syrup, the USA in oil), and both be better off. This is called “**Gains from Specialization**”.
- Canada and the USA could agree on an “exchange rate” of, say 3 barrels of oil per barrel of maple syrup.
- Canada could switch 100,000 barrels of oil production into maple syrup production, and trade that for 300,000 barrels of oil – clearly better off!
- The USA would switch 100,000 barrels of maple syrup production to oil and get 500,000 barrels of oil. It then trades 300,000 of that to Canada to replace the maple syrup and is clearly better off.
- Total production is 2.1M barrels of oil and 500,000 barrels of maple syrup.



Even a country with absolute advantages in production can benefit from trade that makes both parties better off.